

Project Summary

Airline Ticket Reservation System

Keith Wang

I completed this Airline Ticket Reservation System project from database setup to web application implementation. The project was built with Flask, MySQL, PyMySQL, HTML/Jinja templates, and SQL, based on the required airline reservation system schema and use cases.

Database Design & Data Setup

- Implemented and maintained the relational database schema for airlines, airports, airplanes, flights, customers, staff, tickets, and reviews.
- Defined primary keys, foreign keys, compound keys, and constraints to maintain data consistency.
- Created and updated test data for demo scenarios, including airlines, airports, flights, customers, staff accounts, tickets, and reviews.
- Ensured the database could be reset and reloaded cleanly for testing and presentation.

Flask Back-End & Application Logic

- Implemented customer and staff authentication, session handling, and role-based route protection.
- Built Flask routes for customer flight search, ticket purchase, viewing purchased flights, submitting ratings, and checking flight history.
- Built staff routes for viewing flights, creating flights, adding airplanes, changing flight status, viewing ratings, and generating reports.
- Implemented dynamic SQL queries for staff flight filtering by default/current/future/past periods, date range, airport code, and city.
- Used prepared SQL statements through PyMySQL to reduce SQL injection risk.

Business Validation & Error Handling

- Added validation to prevent customers from purchasing departed or cancelled flights.
- Enforced purchase rules such as matching the name on the card with the customer account name and requiring numeric card numbers.
- Added airplane ownership checks so staff can only create flights using airplanes owned by their airline.
- Added duplicate airplane handling and user-friendly flash messages instead of raw database errors.
- Generated safe ticket IDs and handled empty states, invalid input, and authorization errors.

Front-End Templates & User Flow

- Created and updated Jinja/HTML templates for customer and staff dashboards.
- Designed forms, tables, navigation links, filter controls, and feedback messages for each use case.
- Improved staff flight views so the default page shows the next 30 days of flights.
- Improved the create flight workflow so staff can create a flight and still view relevant upcoming flights afterward.
- Organized customer and staff pages to make the demo flow clear and easy to follow.

Testing & Documentation

- Tested the full customer flow: search flights -> purchase ticket -> view my flights -> submit rating/comment.
- Tested the full staff flow: view flights -> add airplane -> create flight -> change status -> view ratings/reports.
- Checked edge cases involving expired sessions, invalid purchases, duplicate records, empty results, and role restrictions.

- Wrote and maintained project documentation, including setup instructions and use-case-to-SQL mapping.

Summary

I independently completed the full-stack implementation of the Airline Ticket Reservation System. My work covered database schema setup, seed data, Flask back-end routes, SQL queries, authentication/session logic, customer and staff features, validation rules, front-end templates, testing, and documentation.